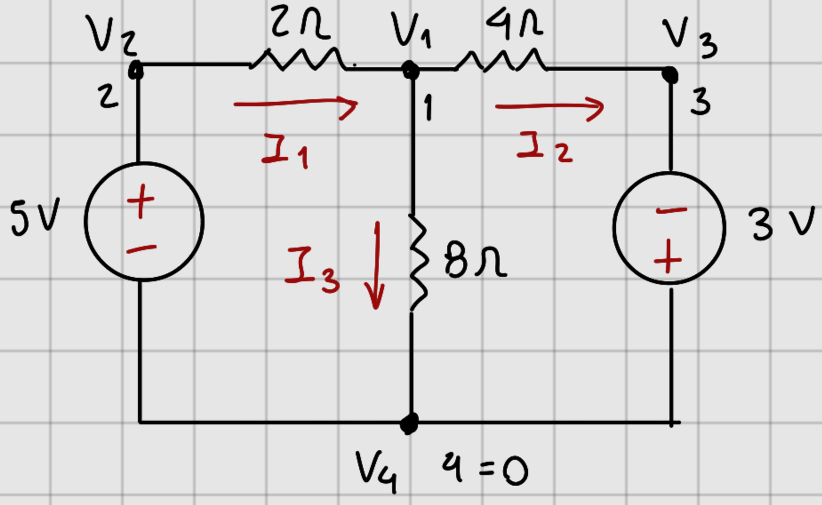
1) Empleando método de nodos determine:



$I_1 = ?$

$I_2 = ?$

$I_3 = ?$

Nodo 1:

$$\frac{V_1 - V_2}{2\Omega} + \frac{V_1}{8\Omega} + \frac{V_1 - V_3}{4\Omega} = 0$$

Nodo 2 :

$$V_2 = 5V$$

Nodo 3:

$$V_3 = -3V$$

Nodo 4:

$$V_4 = 0$$

Nodo 1:

$$\frac{4V_1 - 4V_2 + V_1 + 2V_1 - 2V_3}{8\Omega}$$

$$\frac{7V_1 - 4V_2 - 2V_3}{8\Omega} = 0$$

$$7V_1 - 4V_2 - 2V_3 = 0$$

$$7V_1 - 4(5V) - 2(-3V) = 0$$

$$7V_1 - 20V + 6V = 0$$

$$7V_1 - 14V = 0$$

$$V_1 = \frac{14V}{7} = 2V$$

$$I_1 = \frac{5 - V_1}{2\Omega} = \frac{5V - 2V}{2\Omega} = \frac{3V}{2\Omega}$$

$$I_2 = \frac{3V + 2V}{4\Omega} = \frac{5V}{4\Omega}$$

$$I_1 = 1.5A$$

$$I_2 = 1.25A$$

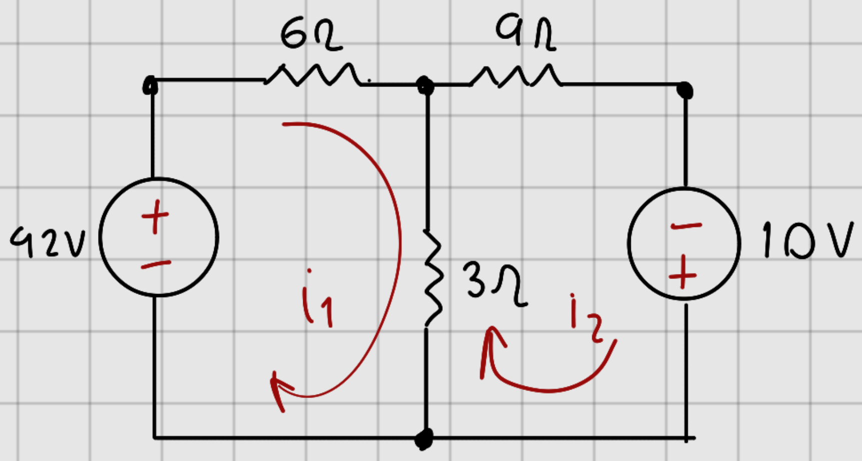
$$I_3 = \frac{V_1}{R} = \frac{2V}{8\Omega} = 0.25A$$

$$I_1 = 1.5A$$

$$I_2 = 1.25A$$

$$I_3 = 0.25A$$

② Resuelva por Mallas



$i_1 = ?$

$i_2 = ?$

Malla 1:

$$6I_1 + 3(I_1 - I_2) = 42V$$

$$6I_1 + 3I_1 - 3I_2 = 42V$$

$$9I_1 - 3I_2 = 42V$$

$$-3I_1 + 7I_2 = 10V$$

Malla 2:

$$3(I_2 - I_1) + 4I_2 = 10V$$

$$3I_2 - 3I_1 + 4I_2 = 10V$$

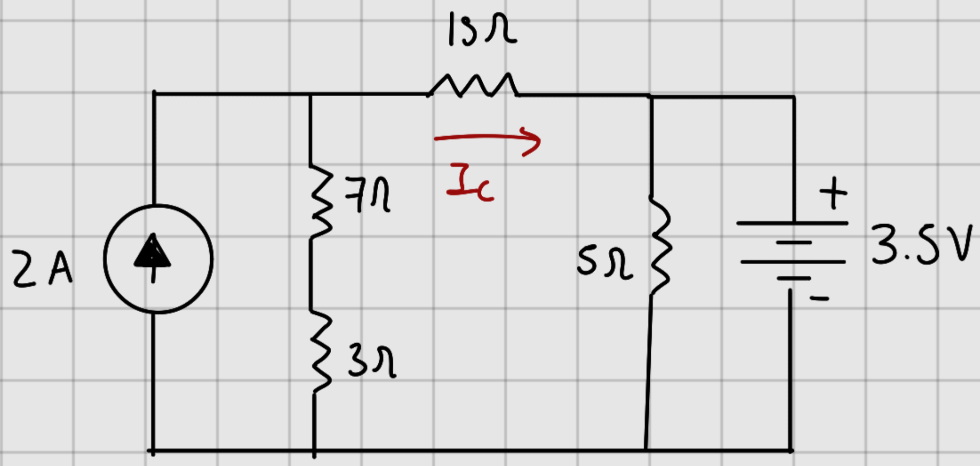
$$7I_2 - 3I_1 = 10V$$

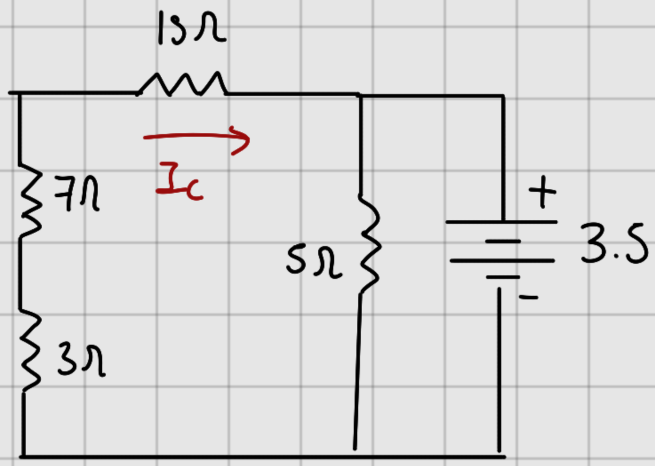
$$\left[\begin{array}{cc|c} 9 & -3 & 42 \\ -3 & 7 & 10 \end{array}\right] \longrightarrow \left[\begin{array}{cc|c} 3 & -1 & 14 \\ 3 & 7 & 10 \end{array}\right] \longrightarrow \left[\begin{array}{cc|c} 3 & -1 & 14 \\ 18 & 0 & 108 \end{array}\right]$$

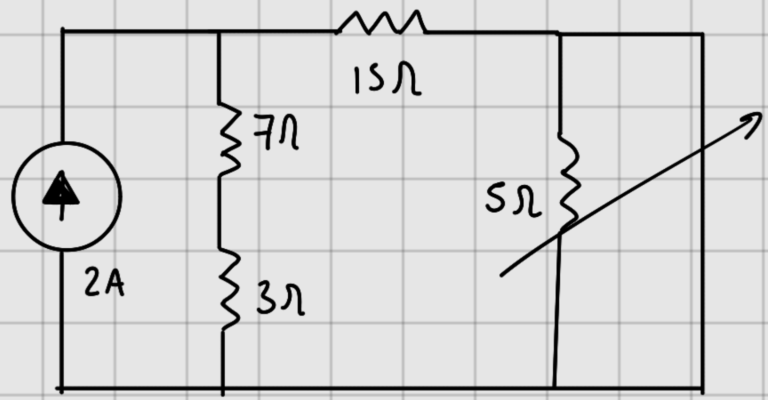
$$\left[\begin{array}{cc|c} 3 & -1 & 14 \\ 1 & 0 & 6 \end{array}\right] \longrightarrow \left[\begin{array}{cc|c} 0 & -1 & -4 \\ 1 & 0 & 6 \end{array}\right] \longrightarrow \left[\begin{array}{cc|c} 0 & 1 & 4 \\ 1 & 0 & 6 \end{array}\right]$$

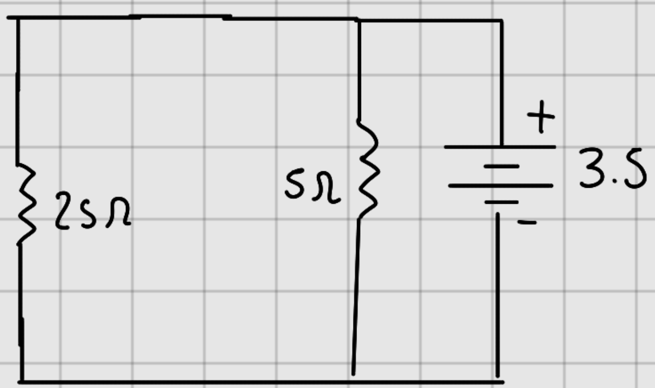
$I_1 = 6A$, $I_2 = 4A$

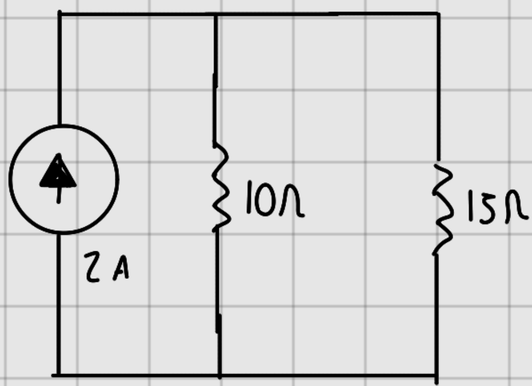
3 Por superposición calcular I_c











$$R_T = \frac{1}{\frac{1}{25\Omega} + \frac{1}{5\Omega}} = \frac{1}{0.04+0.2}$$

$$= \frac{1}{0.24} = 4.166\,\Omega$$

$$R_T = \frac{1}{\frac{1}{10} + \frac{1}{15}} = \frac{1}{0.1+0.0666}$$

$$R_T = \frac{1}{0.1666} = 6\,\Omega$$

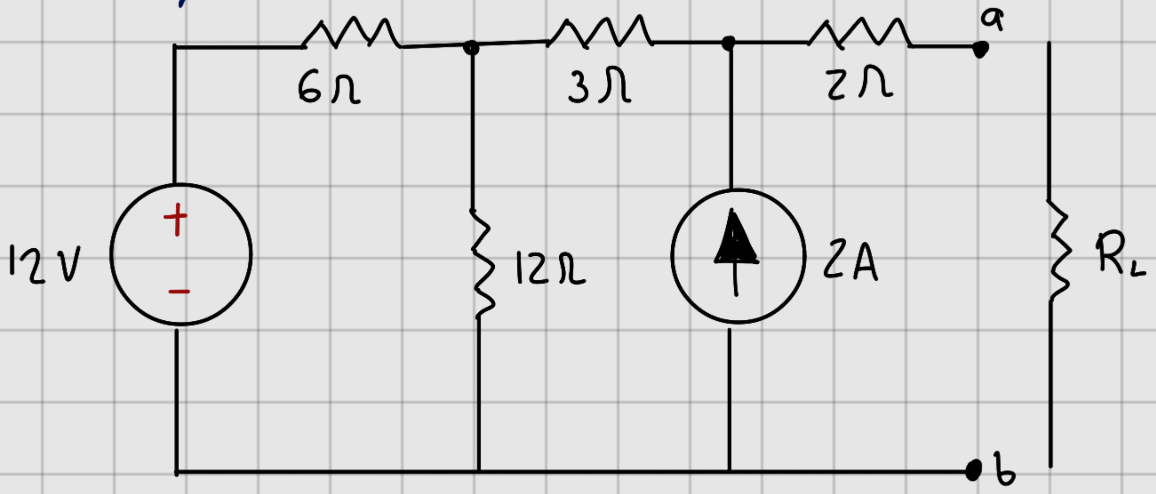
$$I_T = \frac{V}{R} = \frac{3.5\,V}{4.166\,\Omega} = 0.84A \qquad V_T = I_T R_T = 6(2) = 12V$$

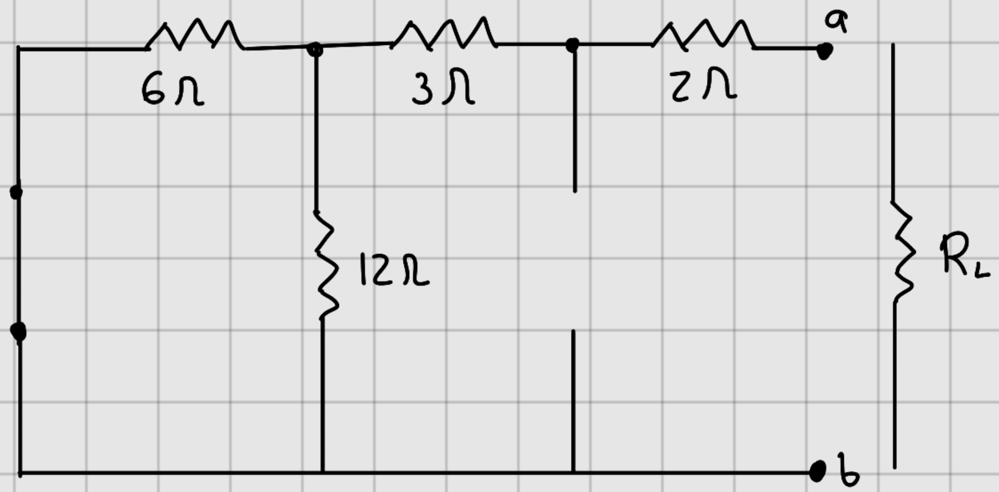
$$I_{25} = \frac{V}{R} = \frac{3.5V}{25\Omega} = 0.14A \qquad I_{15} = \frac{V_T}{R} = \frac{12V}{15\Omega} = 0.8A$$

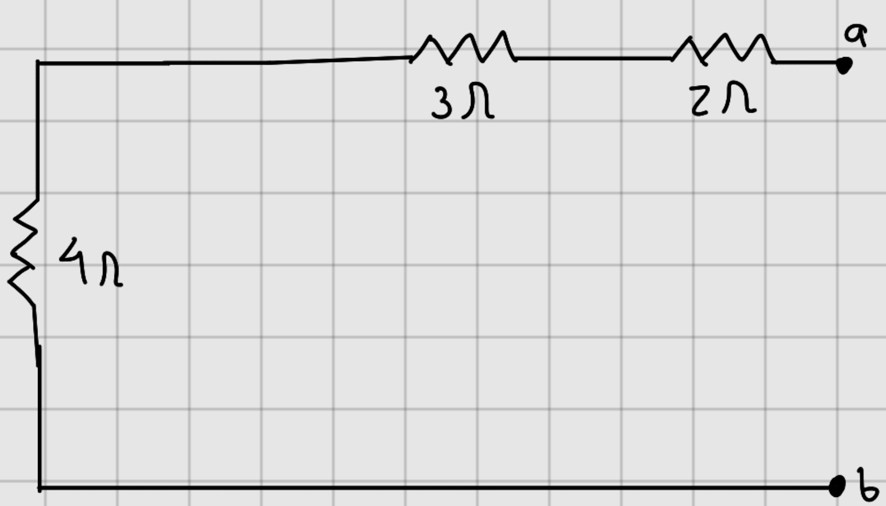
$$I_c = 0.8A - 0.14A = 0.66A$$

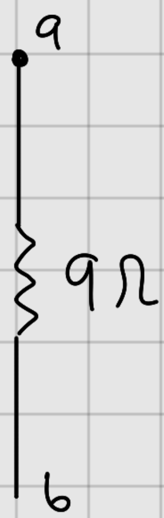
$$I_c = 0.66A$$

④ Calcule V_{TH} y R_{TH} entre los puntos a y b



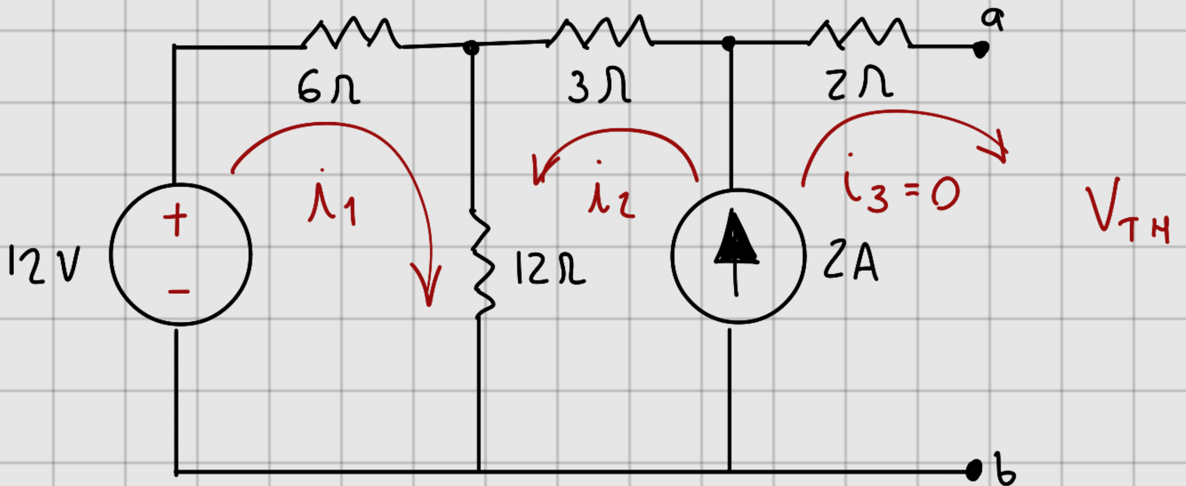






$\underline{R_{TH} = 9\Omega}$

V_{TH}



$$12V - 6i_1 - 12(i_1 + i_2) = 0$$

$$12V - 6i_1 - 12i_1 - 12i_2 = 0$$

$$12V - 18i_1 - 12i_2 = 0$$

$$18i_1 + 12i_2 = -12V$$

$$2A - 3i_2 - 12(i_2 + i_1) + 2\cancel{(0)}^{0} + V_{TH} = 0$$

$$2A - 3i_2 - 12i_2 - 12i_1 = 0$$

$$-15i_2 - 12i_1 + V_{TM} = 0$$

$$2A = i_2 - \cancel{i_3}^{0}$$

$$i_2 = 2A$$

$$18i_1 + 12(2) = 12V$$

$$18i_1 + 24 = 12V$$

$$18i_1 = 12V - 24V$$

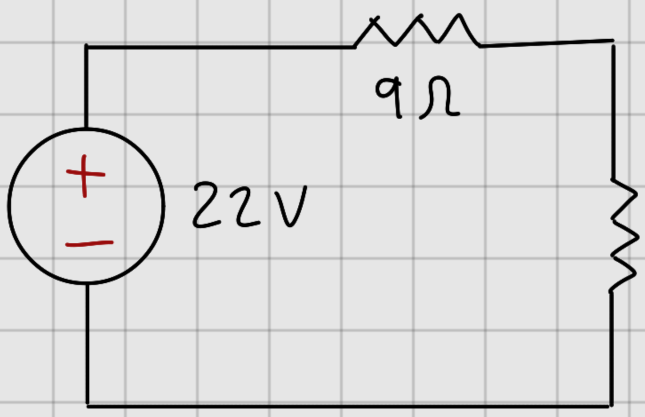
$$i_1 = \frac{-12}{18} = -0.\overline{6}A$$

$$-15(2) - 2(-0.66) + V_{TH} = 0$$

$$-30V + 8V + V_{TH} = 0$$

$$-22V + V_{TH} = 0$$

$$\underline{V_{TH} = 22V}$$



$R_L = 9\Omega$ para máxima transferencia de energía